

Problem Set 6

Mathematics for Social Scientists

EXERCISE 1. Let $\mathbf{u} = (1, 2)$, $\mathbf{v} = (0, 1)$, $\mathbf{w} = (1, -3)$, $\mathbf{x} = (1, 2, 0)$ and $\mathbf{z} = (0, 1, 1)$. Compute each expression (write “not defined” if sizes are incompatible):

- | | |
|---------------------------------|-----------------------------------|
| (i) $\mathbf{u} + \mathbf{v}$ | (vi) $\mathbf{u} + 2\mathbf{v}$ |
| (ii) $-4\mathbf{w}$ | (vii) $\mathbf{u} - \mathbf{v}$ |
| (iii) $\mathbf{x} + \mathbf{z}$ | (viii) $3\mathbf{x} + \mathbf{z}$ |
| (iv) $3\mathbf{z}$ | (ix) $-2\mathbf{x}$ |
| (v) $2\mathbf{v}$ | (x) $\mathbf{w} + 2\mathbf{x}$ |

EXERCISE 2. Find the length (Euclidean norm) of each vector and (optionally) sketch:

- (a) $(3, 4)$ (b) $(0, -3)$ (c) $(1, 1, 1)$ (d) $(3, 3)$ (e) $(-1, -1)$ (f) $(1, 2, 3)$ (g) $(2, 0)$.

EXERCISE 3. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$. If possible, compute $A + B$ and $A \times B$ for the following matrices B .

- (i) $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$
- (ii) $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- (iii) $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$
- (iv) $B = A^T$

EXERCISE 4. For each of the following matrices A , define a vector $v \neq 0$ such that $A \cdot v = v$. If it doesn't exist, state why.

- (i) $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$
- (ii) $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

$$(iii) \quad A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 1 & -1 & 0 \end{pmatrix}$$

EXERCISE 5. Let $A, B \in \mathbb{R}^{3 \times 3}$ be diagonal matrices. Show that $A \times B$ is diagonal.

EXERCISE 6. Let

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & -1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & -1 \\ 4 & -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Compute each matrix if it is defined:

- | | |
|----------------------|-------------------------|
| (i) $A + B$ | (x) CE |
| (ii) $A - D$ | (xi) $-D$ |
| (iii) $3B$ | (xii) $(CE)^\top$ |
| (iv) DC | (xiii) $B + C$ |
| (v) $B^\top$ | (xiv) $D - C$ |
| (vi) $A^\top C^\top$ | (xv) CA |
| (vii) $C + D$ | (xvi) EC |
| (viii) $B - A$ | (xvii) $(CA)^\top$ |
| (ix) AB | (xviii) $E^\top C^\top$ |